Richard Stiskalek, Curriculum Vitae

PROFILE

Affiliation: University of Oxford **Email:** richard.stiskalek@physics.ox.ac.uk
NASA ADS: 17 first-author, 16 co-author papers; h-index = 11; citations = 736
Website: richard-sti.github.io

I am a Hintze Prize Fellow in Astrophysics (Cosmology) and specialise in Bayesian field-level inference and digital twins of the local Universe to bridge observations and theory. My research combines constrained cosmological simulations, semi-analytic galaxy formation models, simulation-based inference, and machine learning to test galaxy physics, map large-scale flows, and deliver precision measurements of cosmological parameters.

EDUCATION

University of Oxford (Balliol College), DPhil in Astrophysics — 2022 – 2026
Thesis: "Constrained Simulations of the Local Universe as a Laboratory for Precision Cosmology and Astrophysics"
Supervised by *Harry Desmond*, *Julien Devriendt*, and *Adrianne Slyz*

Ludwig-Maximilians-Universität München, MSc in Physics — 2020 – 2022
Thesis: "Strong-field lensing of gravitational waves" (partially with Albert Einstein Institute Potsdam)
Supervised by *Miguel Zumalacárregui*, *Marius A. Oancea*, and *Jochen Weller*

Hong Kong University of Science and Technology — 2017 – 2018
Undergraduate Exchange Programme in Physics

University of Glasgow, BSc in Physics with Astrophysics — 2016 – 2020
Thesis: "Gravitational-wave cosmology"
Supervised by *Martin Hendry*

ACADEMIC APPOINTMENTS

University of Oxford, Hintze Centre for Astrophysical Surveys — 2026 – 2029
Hintze Prize Fellow in Astrophysics (Cosmology)

Flatiron Institute, Center for Computational Astrophysics — 2025
Pre-Doctoral Fellow (with *Shy Genel* and *Lucia A. Perez*)

Institut d'astrophysique de Paris — 2024
Long Term Attachment (with *Guilhem Lavaux*)

Max Planck Institute for Gravitational Physics, Observational Relativity and Cosmology — 2020
Summer Research Intern (with *Collin Capano*)

University of Oxford, Sub-department of Astrophysics — 2019
Summer Research Intern (with *Harry Desmond*)

University of Glasgow, Institute for Gravitational Research — 2018
Summer Research Intern (with *John Veitch* and *Chris Messenger*)

COLLABORATION MEMBERSHIPS

Learning the Universe — 2025 – present
Simons Foundation collaboration developing next-generation cosmological simulations, machine learning methods, and advanced Bayesian inference for large-scale surveys
Contributions: Training set generation and BORG team

CAMELS — 2025 – present
Collaboration building large public suites of cosmological hydrodynamical and N-body simulations spanning broad cosmological and astrophysical parameter ranges to train and test machine learning methods
Contributions: Leading projects on semi-analytic modelling and simulation-based inference

Aquila Consortium — 2022 – present
International collaboration developing Bayesian field-level inference techniques to reconstruct initial conditions and study large-scale structure formation
Contributions: Leading projects on data generation, benchmarking, large-scale flows and cosmological parameters

Teaching, Mentoring & Service

Student Supervision

Noah Pierce (BSc, Southampton; with H. Desmond & T. Yasin) 2025 – present
"Field-Level Inference of the Galaxy Luminosity Function"

Joshua Darne (MPhys, Oxford; with T. Yasin & H. Desmond) 2024
"Radial acceleration relation in the NewHorizon hydrodynamical simulation"

Fedir Boreiko (BSc, Manchester; with T. Yasin & H. Desmond) 2024
"The correlation between light and dark matter across cosmic time" [arXiv:2601.07799]

Enoch Ko (BSc, Warwick; with T. Yasin & H. Desmond) 2024
"Dark matter and galaxy dynamics: enduring puzzles" [arXiv:2508.03569]

Catherine Gallagher (MPhys, Oxford; with T. Yasin & H. Desmond) 2023 – 2024
"The influence of cosmic environment on galaxy properties" [arXiv:2503.14732]

James Harvey (BSc, Oxford; with T. Yasin & H. Desmond) 2023 – 2024
"Machine learning the time of last major merger from spectroscopic data"

Teaching Experience

Tutor, *MPhys C1 Astrophysics*, University of Oxford 2023 – 2024
Graduate tutorials in cosmology and stellar astrophysics

Mentor, *Lumiere Education* 2023 – present
Supervision of senior high school students carrying out independent research projects

Demonstrator, *3rd-Year Practical Course*, University of Oxford 2023
Computational astrophysics practical sessions

Academic Service

Referee for *A&A*, *ApJ*, *MNRAS*, *OJA*, *PNAS*, and *PRD* 2022 – present
Reviewed 20+ manuscripts for leading journals

Coordinator, Aquila Consortium Monthly Telecon 2023 – present

Organiser, "Simulator's Coffee", University of Oxford 2023 – 2024

Local organiser, Aquila Consortium Oxford Meeting 2023

Lead organiser, "Middle of Scotland Science Festival" 2018

Selected Awards and Scholarships

Hintze Prize Fellowship in Astrophysics (Cosmology), University of Oxford 2026
Flatiron Research Fellowship, Flatiron Institute (declined) 2026
Junior Research Fellowship, Emmanuel College, Cambridge (declined) 2026
Snell Exhibition, Balliol College 2022 – 2026
STFC PhD Funding, Science and Technology Facilities Council 2022 – 2026
DAAD Study Scholarship, German Academic Exchange Service 2021 – 2022
Kerr Bursary, University of Glasgow 2020
Lang Scholarship, University of Glasgow 2019
Undergraduate Summer Bursary, Royal Astronomical Society 2018
Dean's List, Hong Kong University of Science and Technology 2018
Astronomy 1 Prize, University of Glasgow 2017
Matthew A. Muir Bursary, University of Glasgow 2017
South East Asia Study Abroad Scholarship, University of Glasgow 2017 – 2018

Skills

Programming: Python, Julia, C/C++, Fortran; high-performance computing (MPI, GPU acceleration)

Statistics & Machine Learning: Bayesian inference (NumPyro/JAX, PyMC, Stan), simulation-based inference, density estimation; deep learning (MLPs, CNNs, GNNs, normalising flows) with JAX, TensorFlow, PyTorch; tree-based methods (random forests, gradient boosting) with scikit-learn

Astrophysics Software: RAMSES, AREPO, Gadget, Rockstar, DisPerSe; N-body/hydrodynamical simulation analysis

Languages: English (fluent), Czech (native), Slovak (fluent), French (intermediate), German (beginner)

PUBLICATIONS

First-Author – Submitted

[1] "Learning the Universe with cosmological rescaling of merger trees and semi-analytic galaxy formation models"; **R. Stiskalek**, L. A. Perez, S. Genel, R. S. Somerville, R. E. Angulo, S. Contreras [arXiv:2606.10024]

[2] "Forward-modelling Milky Way Cepheids: selection effects and physical priors in the Gaia–HST calibration"; **R. Stiskalek**, A. Riess, H. Desmond, G. Lavaux, D. Scolnic [arXiv:2603.09880]

[3] "S_8 from Tully–Fisher, fundamental plane, and supernova distances agree with *Planck*"; **R. Stiskalek** [arXiv:2509.20235]

First-Author – Accepted

[4] "Validating Digital Twins of the Local Universe with the Thermal Sunyaev-Zel'dovich Signal"; **R. Stiskalek**, H. Desmond [Open Journal of Astrophysics 9, arXiv:2601.15935]

[5] "Revisiting the Great Attractor: The Local Group's streamline trajectory, cosmic velocity and dynamical fate"; **R. Stiskalek**, H. Desmond, S. McAlpine, G. Lavaux, J. Jasche, M. J. Hudson [Open Journal of Astrophysics 9, arXiv:2601.08524]

[6] "1.8 per cent measurement of H_0 from Cepheids alone"; **R. Stiskalek**, H. Desmond, E. Tsaprazi, A. Heavens, G. Lavaux, S. McAlpine, J. Jasche [MNRAS 546:staf2260, arXiv:2509.09665]

[7] "No evidence for local H_0 anisotropy from Tully–Fisher or supernova distances"; **R. Stiskalek**, H. Desmond, G. Lavaux [MNRAS 546:staf2048, arXiv:2509.14997]

[8] "The Velocity Field Olympics: Assessing velocity field reconstructions with direct distance tracers"; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz, G. Lavaux, M. J. Hudson, D. J. Bartlett, H. Courtois [MNRAS 545:staf1960, arXiv:2502.00121]

[9] "Testing the local supervoid solution to the Hubble tension with direct distance tracers"; **R. Stiskalek**, H. Desmond, I. Banik [MNRAS 543:1556, arXiv:2506.10518]

[10] "Symmetry in Hyper Suprime-Cam galaxy spin directions"; **R. Stiskalek**, H. Desmond [Res. Notes AAS 8 281, arXiv:2410.18884]

[11] "Evaluating the variance of individual halo properties in constrained cosmological simulations"; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz [MNRAS 534:3120, arXiv:2310.20672]

[12] "Probing general relativistic spin-orbit coupling with gravitational waves from hierarchical triple systems"; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [MNRAS 535:L1, arXiv:2307.01903]

[13] "On the fundamentality of the radial acceleration relation for late-type galaxy dynamics"; **R. Stiskalek**, H. Desmond [MNRAS 525:6130, arXiv:2305.19978]

[14] "Frequency- and polarization-dependent lensing of gravitational waves in strong gravitational fields"; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [Phys. Rev. D 109, 124045, arXiv:2209.06459]

[15] "The scatter in the galaxy–halo connection: a machine learning analysis"; **R. Stiskalek**, D. J. Bartlett, H. Desmond, D. Anbajagane [MNRAS 514:4026, arXiv:2202.14006]

[16] "The dependence of subhalo abundance matching on galaxy photometry and selection criteria"; **R. Stiskalek**, H. Desmond, T. Holvey, M. G. Jones [MNRAS 506:3205, arXiv:2101.02765]

[17] "Are stellar-mass binary black hole mergers isotropically distributed?"; **R. Stiskalek**, J. Veitch, C. Messenger [MNRAS 501:970, arXiv:2003.02919]

Co-Author – Submitted

[1] "Learning the Universe: Constrained simulations of the Coma galaxy cluster – I. Radial X-ray and Compton-y signatures"; U. P. Steinwandel, S. McAlpine, **R. Stiskalek**,

Rüdiger Pakmor, V. Springel, E. Churazov, I. Khabibullin, J. Jasche, G. Lavaux, G. L. Bryan [arXiv:2606.10028]

[2] "Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation"; V. Pandya, . . . , **R. Stiskalek**, . . . [arXiv:2604.06318]

[3] "Parameterizing Dark Energy at the density level: A two-parameter alternative to CPL"; G. Montefalcone, **R. Stiskalek** [arXiv:2603.25735]

[4] "Testing cosmic anisotropy with cluster scaling relations"; T. Yasin, **R. Stiskalek**, H. Desmond, S. von Hausegger, P. G. Ferreira [arXiv:2602.06007]

[5] "Testing subhalo abundance matching with galaxy kinematics"; F. Boreiko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [arXiv:2601.07799]

[6] "MEGATRON: Reproducing the Diversity of High-Redshift Galaxy Spectra with Cosmological Radiation Hydrodynamics Simulations"; H. Katz, . . . , **R. Stiskalek**, . . . [arXiv:2510.05201]

[7] "MEGATRON: how the first stars create an iron metallicity plateau in the smallest dwarf galaxies"; M. Rey, . . . , **R. Stiskalek**, . . . [arXiv:2510.05232]

[8] "MEGATRON: the impact of non-equilibrium effects and local radiation fields on the circumgalactic medium at cosmic noon"; C. Cadiou, . . . , **R. Stiskalek**, . . . [arXiv:2510.05667]

Co-Author – Accepted

[9] "Constraining cosmological simulations with peculiar velocities: a forward-modeling approach"; A. Valade, N. Libeskind, D. Pomarède, **R. Stiskalek**, Y. Hoffman, S. Gottlöber, R. B. Tully [A&A, in press, arXiv:2602.03699]

[10] "The subtle statistics of the distance ladder: On the distance prior and selection effects"; H. Desmond, **R. Stiskalek**, J. A. Najera, I. Banik [in press, arXiv:2511.03394]

[11] "The galaxy-environment connection revealed by constrained simulations"; C. Gallagher, T. Yasin, **R. Stiskalek**, H. Desmond, M. J. Jarvis [MNRAS 546:stag108, arXiv:2503.14732]

[12] "Renzo's rule revisited: A statistical study of galaxies' baryon–dark matter coupling"; E. Ko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [MNRAS 544:4288, arXiv:2508.03569]

[13] "CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning"; N. Huang, **R. Stiskalek**, J.-Y. Lee, A. E. Bayer, C. C. Margossian, C. K. Jespersen, L. A. Perez, L. K. Saul, F. Villaescusa-Navarro [NeurIPS 2025, arXiv:2507.03707]

[14] "The Manticore Project I: a digital twin of our cosmic neighbourhood from Bayesian field-level analysis"; S. McAlpine, J. Jasche, M. Ata, G. Lavaux, **R. Stiskalek**, C. S. Frenk, A. Jenkins [MNRAS 540:716, arXiv:2505.10682]

[15] "The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics"; Eleonora Di Valentino, . . . , **R. Stiskalek**, . . . [Physics of the Dark Universe, Volume 49, 2025, 101965, arXiv:2504.01669]

[16] "Inferring the Ionizing Photon Contributions of High-Redshift Galaxies to Reionization with JWST NIRCam Photometry"; N. Choustikov, **R. Stiskalek**, A. Saxena, H. Katz, J. Devriendt, A. Slyz [MNRAS 537:2273, arXiv:2405.09720]

INVITED TALKS

Supernova-free local ladders for H_0 **2026**

GDR CoPhy Episode 4, Clermont-Ferrand, France
Cosmology seminar, UCL

Bridging Machine Learning and Cosmology at Scale

PV Reconstruction online meeting

Is the Hubble tension easing?

University of Oxford

Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling
CAMELS + Baryon Pasting Workshop, University of Osaka, Japan

Velocity Field Olympics
Leibniz Institute for Astrophysics Potsdam, Germany

Forward modelling of peculiar velocities and large-scale structure **2025**
Institut d'Astrophysique de Paris

Pinning down the local Universe with digital twins
University of Cambridge
Yale University

Do SH0ES Cepheids prefer the Planck Hubble constant?
National Astronomy Meeting, Durham

Towards accurate models and tests of the local Universe
Columbia University
Princeton University
Center for Computational Astrophysics, Flatiron Institute

Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling
Center for Computational Astrophysics, Flatiron Institute

Velocity Field Olympics
Center for Computational Astrophysics, Flatiron Institute
Cosmic Flows 2025, Brisbane

Velocity Field Olympics **2024**
University of Portsmouth

Search for the optimal dark matter halo density profile **2023**
University of Oxford

Is the radial acceleration relation a fundamental correlation?
University of Oxford

Frequency and polarization dependent propagation of gravitational waves **2022**
University of Glasgow
LMU Munich
Max Planck Institute for Gravitational Physics, Potsdam

The scatter in the galaxy–halo connection
Baryon Pasters Collaboration meeting

The scatter in the galaxy–halo connection **2021**
LMU Munich

Reversible-jump MCMC in gravitational-wave astronomy **2020**
Max Planck Institute for Gravitational Physics, Hannover

Are binary-black hole mergers isotropically distributed?
LIGO Scientific Collaboration Data Analysis telecon

The relation between galaxies and dark matter halos **2019**
University of Oxford